AC Sweep 1

Mag: PR1: V(3)-V(4)
Phase: PR1: V(3)-V(4)

Magnitude (Logarithmic)

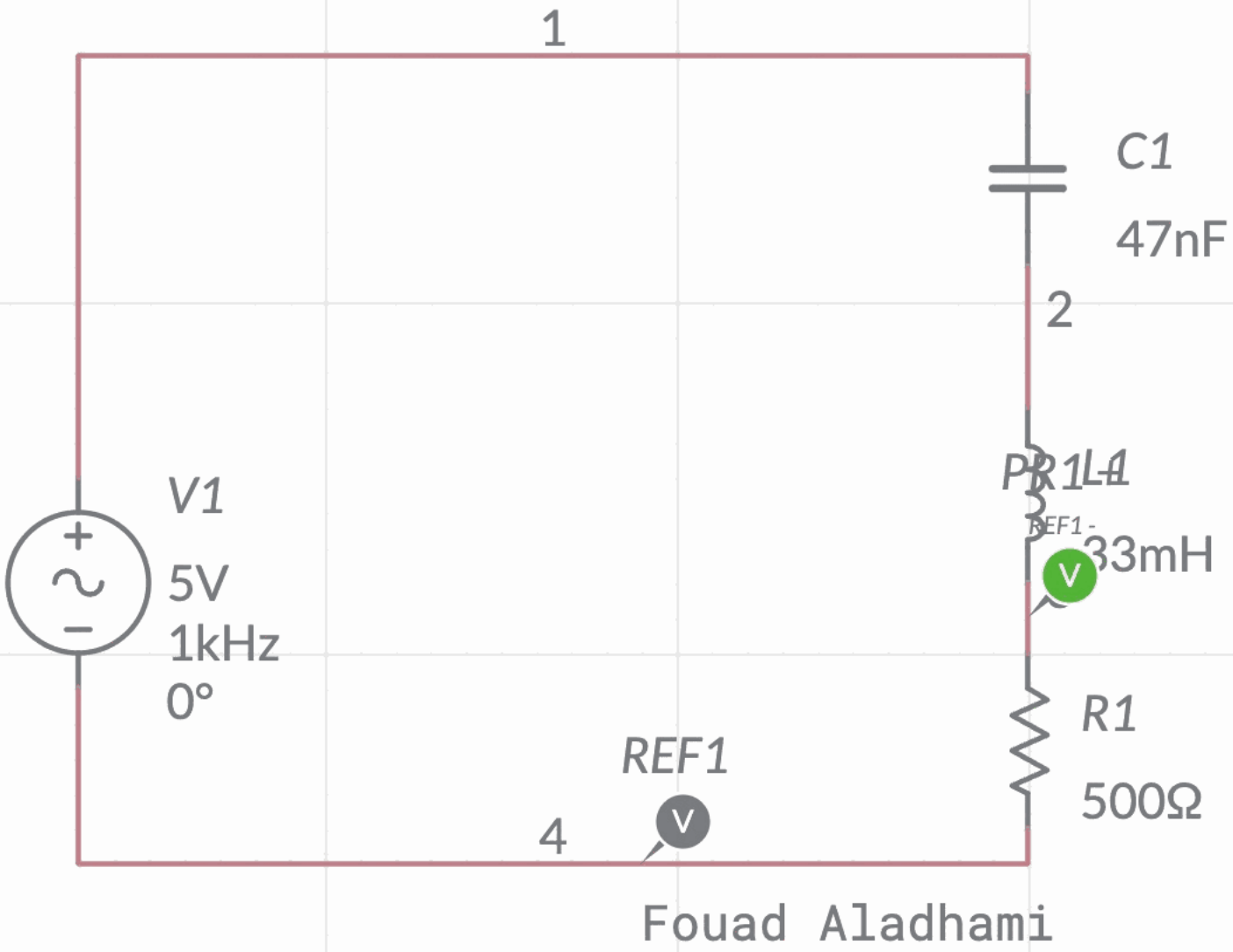
1
100m
10m
1m
100µ
10µ
1µ
100n

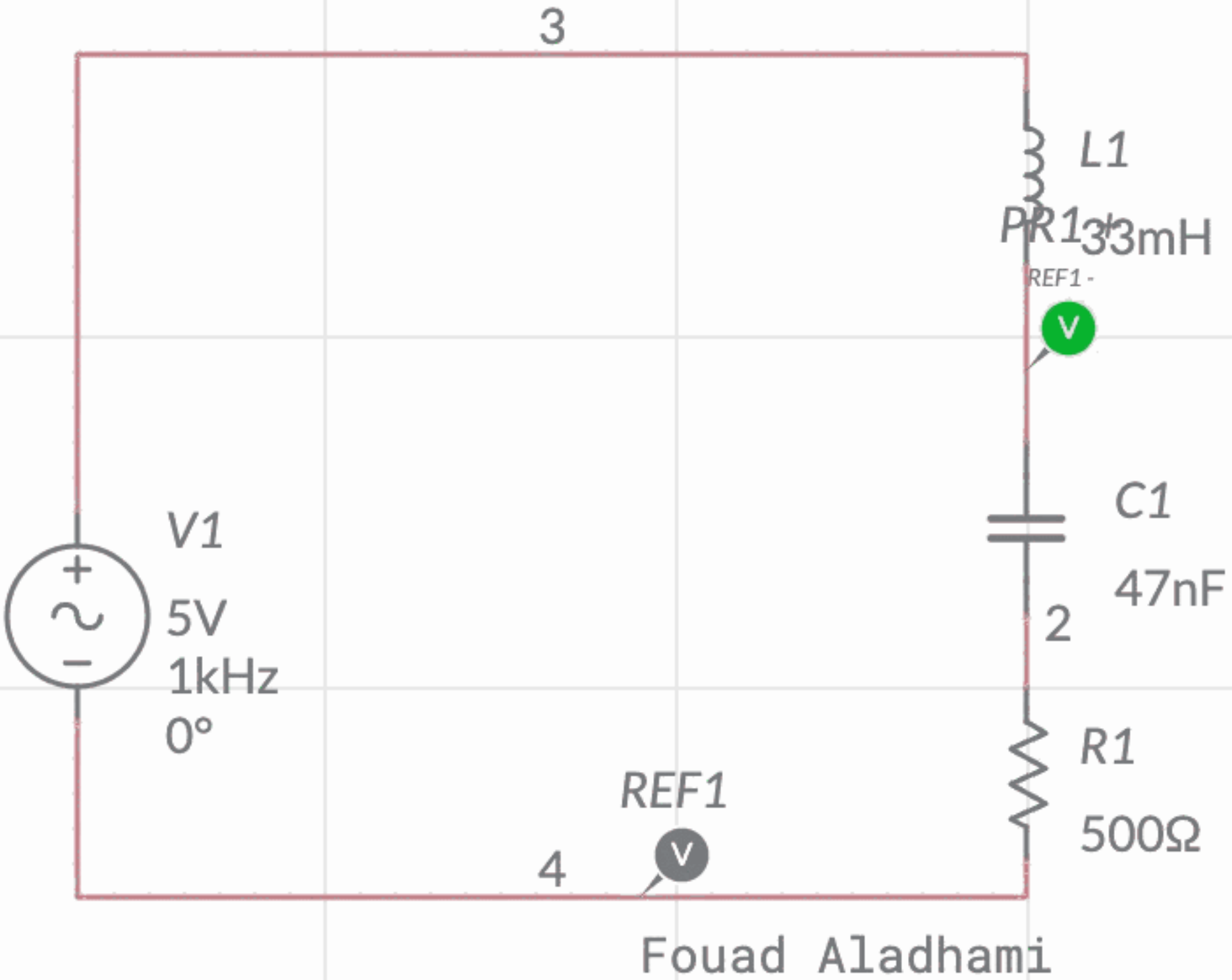
Phase (°)

100
75
50
25
0
-25
-50
-75
-100

1
10
100
1k
10k
100k
1M
10M
100M
1G
10G

Frequency (Hz)


1
C1
47nF
2
V1
5V
1kHz
0°
REF1
4
R1
500Ω
33mH
Fouad Aladhami


3
L1
33mH
REF1 -
V
C1
47nF
2
V1
5V
1kHz
0°
R1
500Ω
REF1
V
4
Fouad Aladhami

AC Sweep 1

Mag: PR1: V(1)-V(4)
Phase: PR1: V(1)-V(4)

Magnitude (Logarithmic)
1
100m
10m
1m
100µ
10µ
1µ
100n

Phase (°)
0
-10
-20
-30
-40
-50
-60
-70
-80
-90
-100
-110
-120

1
10
100
1k
10k
100k
1M
10M
100M
1G
10G
Frequency (Hz)

Figure 1

File Edit View Insert Tools Desktop Window Help

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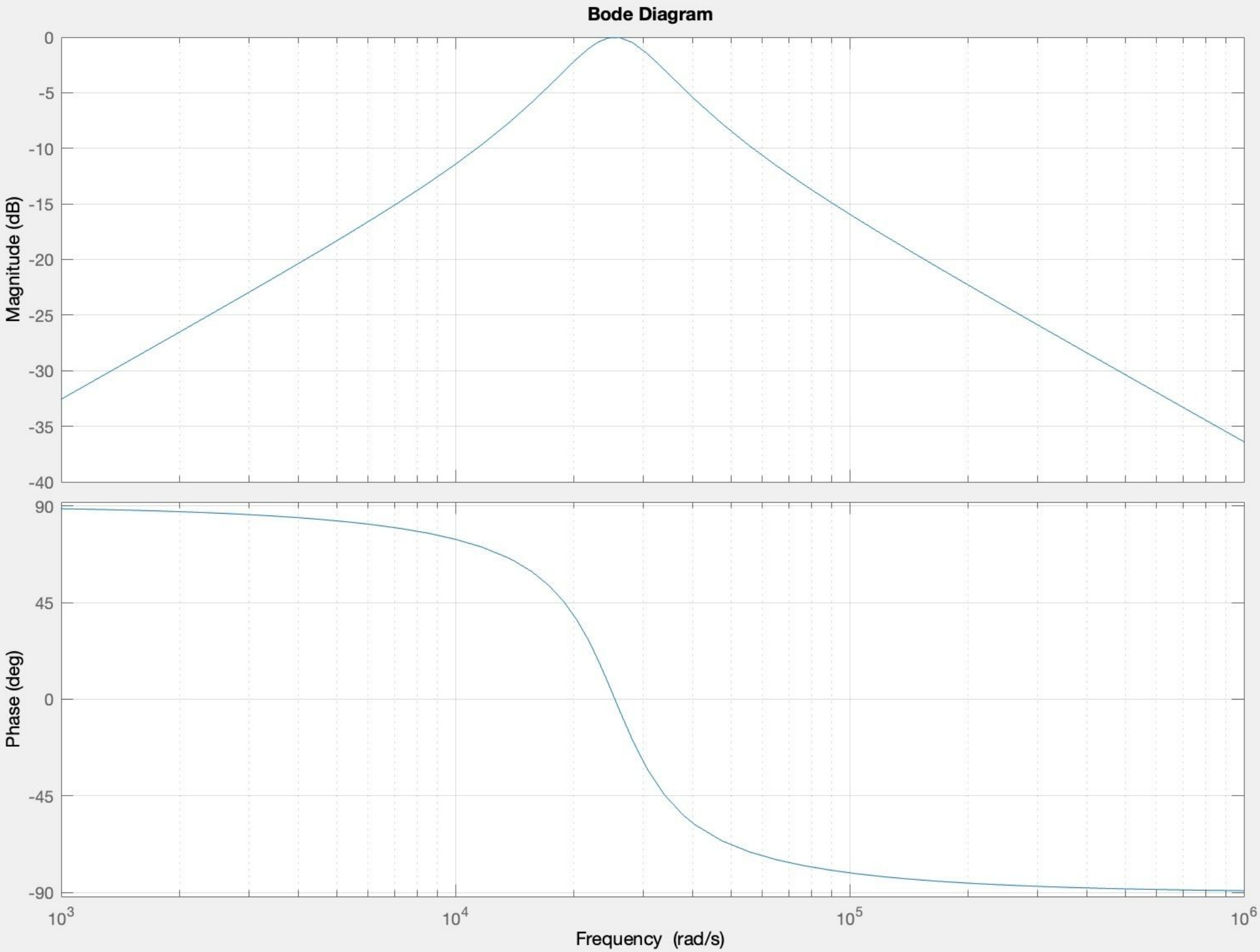
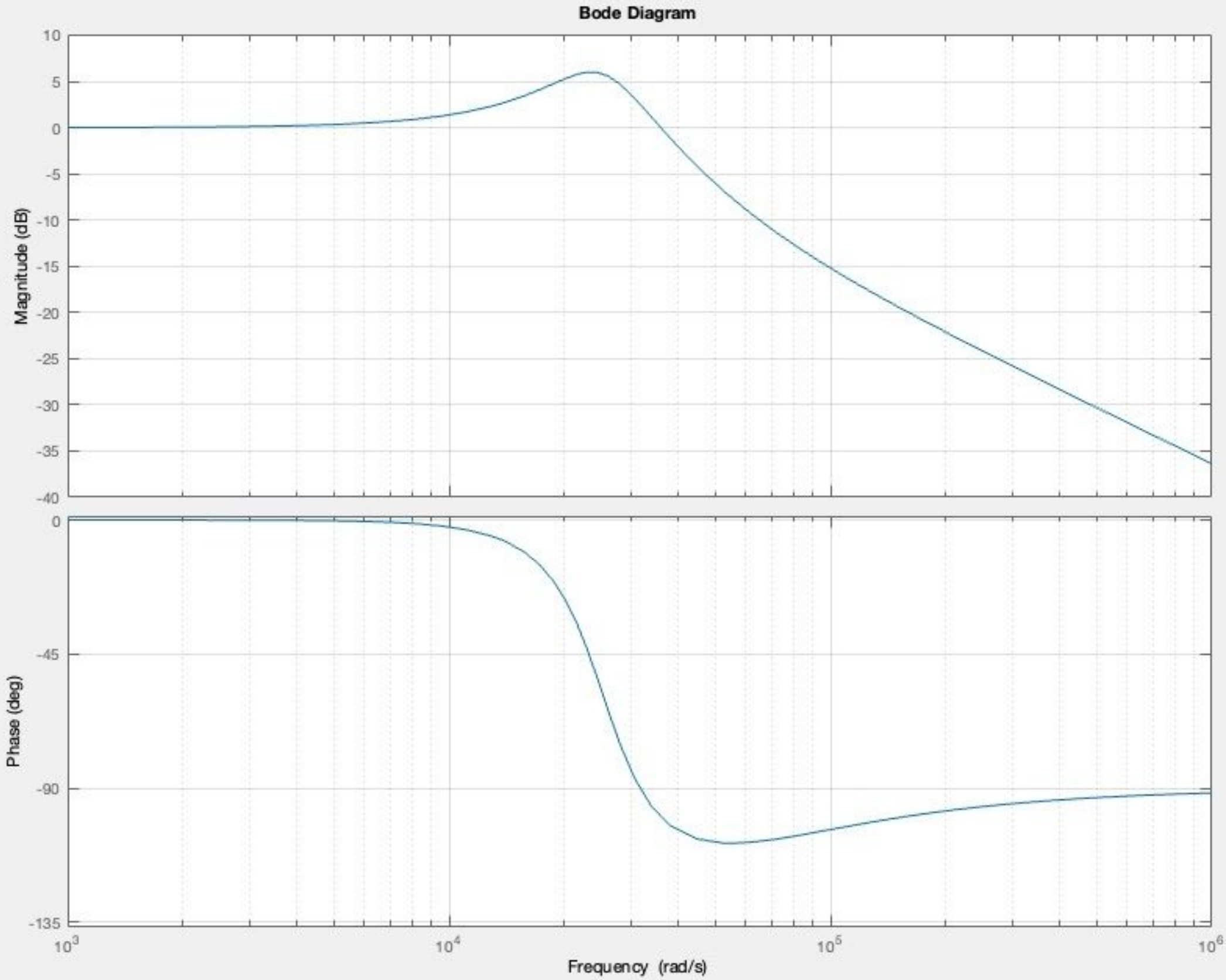


Figure 2

File Edit View Insert Tools Desktop Window Help



```
R= 500;
C = 47.*10.^-9;
L = 33.*10.^-3;
w0  = 1./ sqrt(L*C)
f0 = w0./ (2.*pi)
w1 = - (R./(2*L)) + sqrt( ( (R ./ (2*L)).^2) + 1./(L.*C) )
f1 = w1./ (2.*pi)
w2 =  (R./(2*L)) + sqrt( ( (R ./ (2*L)).^2) + 1./(L.*C) )
f2 = w2./ (2.*pi)
B = w2 - w1
Q = (w0.*L)./R

wc = 1 ./(R.*C)
fc = wc ./ (2*pi)
Hs1 = tf([R,0],[L,R,1/C]);
bode(Hs1)
grid on
figure
Hs2 = tf([R,1/C],[L,R,1/C]);
bode(Hs2)
grid on
```